MeshCore KISS Modem Protocol

Standard KISS TNC firmware for MeshCore LoRa radios. Compatible with any KISS client (Direwolf, APRSdroid, YAAC, etc.) for sending and receiving raw packets. MeshCore-specific extensions (cryptography, radio configuration, telemetry) are available through the standard SetHardware (0x06) command.

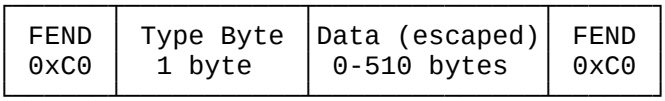
Serial Configuration

115200 baud, 8N1, no flow control.

Frame Format

Standard KISS framing per the KA9Q/K3MC specification.

Byte	Name	Description
`0xC0`	FEND	Frame delimiter
`0xDB`	FESC	Escape character
`0xDC`	TFEND	Escaped FEND (FESC + TFEND = 0xC0)
`0xDD`	TFESC	Escaped FESC (FESC + TFESC = 0xDB)



Type Byte

The type byte is split into two nibbles:

Bits	Field	Description
7-4	Port	Port number (0 for single-port TNC)
3-0	Command	Command number

Maximum unescaped frame size: 512 bytes.

Standard KISS Commands

Host to TNC

Command	Value	Data	Description
Data	`0x00`	Raw packet	Queue packet for transmission
TXDELAY	`0x01`	Delay (1 byte)	Transmitter keyup delay in 10ms units (default: 50 = 500ms)
Persistence	`0x02`	P (1 byte)	CSMA persistence parameter 0-255 (default: 63)
SlotTime	`0x03`	Interval (1 byte)	CSMA slot interval in 10ms units (default: 10 = 100ms)
TXtail	`0x04`	Delay (1 byte)	Post-TX hold time in 10ms units (default: 0)
FullDuplex	`0x05`	Mode (1 byte)	0 = half duplex, nonzero = full duplex (default: 0)
SetHardware	`0x06`	Sub-command + data	MeshCore extensions (see below)
Return	`0xFF`	-	Exit KISS mode (no-op)

TNC to Host

Type	Value	Data	Description
Data	`0x00`	Raw packet	Received packet from radio

Data frames carry raw packet data only, with no metadata prepended. The Data command payload is limited to 255 bytes to match the MeshCore maximum transmission unit (MAX_TRANS_UNIT); frames larger than 255 bytes are

silently dropped. The KISS specification recommends at least 1024 bytes for general-purpose TNCs; this modem is intended for MeshCore packets only, whose protocol MTU is 255 bytes.

CSMA Behavior

The TNC implements p-persistent CSMA for half-duplex operation:

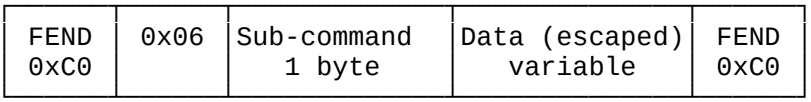
1. When a packet is queued, monitor carrier detect
2. When the channel clears, generate a random value 0-255
3. If the value is less than or equal to P (Persistence), wait TXDELAY then transmit
4. Otherwise, wait SlotTime and repeat from step 1

In full-duplex mode, CSMA is bypassed and packets transmit after TXDELAY.

SetHardware Extensions (0x06)

MeshCore-specific functionality uses the standard KISS SetHardware command. The first byte of SetHardware data is a sub-command. Standard KISS clients ignore these frames.

Frame Format



Request Sub-commands (Host to TNC)

Sub-command	Value	Data
GetIdentity	0x01	-
GetRandom	0x02	Length (1 byte, 1-64)
VerifySignature	0x03	PubKey (32) + Signature (64) + Data
SignData	0x04	Data to sign
EncryptData	0x05	Key (32) + Plaintext
DecryptData	0x06	Key (32) + MAC (2) + Ciphertext
KeyExchange	0x07	Remote PubKey (32)
Hash	0x08	Data to hash
SetRadio	0x09	Freq (4) + BW (4) + SF (1) + CR (1)
SetTxPower	0x0A	Power dBm (1)
GetRadio	0x0B	-
GetTxPower	0x0C	-
GetCurrentRssi	0x0D	-
IsChannelBusy	0x0E	-
GetAirtime	0x0F	Packet length (1)
GetNoiseFloor	0x10	-
GetVersion	0x11	-
GetStats	0x12	-
GetBattery	0x13	-
GetMCUTemp	0x14	-
GetSensors	0x15	Permissions (1)
GetDeviceName	0x16	-
Ping	0x17	-
Reboot	0x18	-
SetSignalReport	0x19	Enable (1): 0x00=disable, nonzero=enable

Sub-command	Value	Data
GetSignalReport	`0x1A`	-

Response Sub-commands (TNC to Host)

Response codes use the high-bit convention: `response = command | 0x80`. Generic and unsolicited responses use the `0xF0`+ range.

Sub-command	Value	Data
Identity	`0x81`	PubKey (32)
Random	`0x82`	Random bytes (1-64)
Verify	`0x83`	Result (1): 0x00=invalid, 0x01=valid
Signature	`0x84`	Signature (64)
Encrypted	`0x85`	MAC (2) + Ciphertext
Decrypted	`0x86`	Plaintext
SharedSecret	`0x87`	Shared secret (32)
Hash	`0x88`	SHA-256 hash (32)
Radio	`0x8B`	Freq (4) + BW (4) + SF (1) + CR (1)
TxPower	`0x8C`	Power dBm (1)
CurrentRssi	`0x8D`	RSSI dBm (1, signed)
ChannelBusy	`0x8E`	Result (1): 0x00=clear, 0x01=busy
Airtime	`0x8F`	Milliseconds (4)
NoiseFloor	`0x90`	dBm (2, signed)
Version	`0x91`	Version (1) + Reserved (1)
Stats	`0x92`	RX (4) + TX (4) + Errors (4)
Battery	`0x93`	Millivolts (2)
MCUTemp	`0x94`	Temperature (2, signed)
Sensors	`0x95`	CayenneLPP payload
DeviceName	`0x96`	Name (variable, UTF-8)
Pong	`0x97`	-
SignalReport	`0x9A`	Status (1): 0x00=disabled, 0x01=enabled
OK	`0xF0`	-
Error	`0xF1`	Error code (1)
TxDone	`0xF8`	Result (1): 0x00=failed, 0x01=success
RxMeta	`0xF9`	SNR (1) + RSSI (1)

Error Codes

Code	Value	Description
InvalidLength	`0x01`	Request data too short
InvalidParam	`0x02`	Invalid parameter value
NoCallback	`0x03`	Feature not available
MacFailed	`0x04`	MAC verification failed
UnknownCmd	`0x05`	Unknown sub-command
EncryptFailed	`0x06`	Encryption failed
TxBusy	`0x07`	Transmit busy

Unsolicited Events

The TNC sends these SetHardware frames without a preceding request:

TxDone (0xF8): Sent after a packet has been transmitted. Contains a single byte: 0x01 for success, 0x00 for failure.

RxMeta (0xF9): Sent immediately after each standard data frame (type 0x00) with metadata for the received packet. Contains SNR (1 byte, signed, value x4 for 0.25 dB precision) followed by RSSI (1 byte, signed, dBm). Enabled by default; can be toggled with SetSignalReport. Standard KISS clients ignore this frame.

Data Formats

Radio Parameters (SetRadio / Radio response)

All values little-endian.

Field	Size	Description
Frequency	4 bytes	Hz (e.g., 869618000)
Bandwidth	4 bytes	Hz (e.g., 62500)
SF	1 byte	Spreading factor (5-12)
CR	1 byte	Coding rate (5-8)

Version (Version response)

Field	Size	Description
Version	1 byte	Firmware version
Reserved	1 byte	Always 0

Encrypted (Encrypted response)

Field	Size	Description
MAC	2 bytes	HMAC-SHA256 truncated to 2 bytes
Ciphertext	variable	AES-128 block-encrypted data with zero padding

Airtime (Airtime response)

All values little-endian.

Field	Size	Description
Airtime	4 bytes	uint32_t, estimated air time in milliseconds

Noise Floor (NoiseFloor response)

All values little-endian.

Field	Size	Description
Noise floor	2 bytes	int16_t, dBm (signed)

The modem recalibrates the noise floor every 2 seconds with an AGC reset every 30 seconds.

Stats (Stats response)

All values little-endian.

Field	Size	Description
RX	4 bytes	Packets received
TX	4 bytes	Packets transmitted
Errors	4 bytes	Receive errors

Battery (Battery response)

All values little-endian.

Field	Size	Description
Millivolts	2 bytes	uint16_t, battery voltage in mV

MCU Temperature (MCUTemp response)

All values little-endian.

Field	Size	Description
Temperature	2 bytes	int16_t, tenths of °C (e.g., 253 = 25.3°C)

Returns `NoCallback` error if the board does not support temperature readings.

Device Name (DeviceName response)

Field	Size	Description
Name	variable	UTF-8 string, no null terminator

Reboot

Sends an `OK` response, flushes serial, then reboots the device. The host should expect the connection to drop.

Sensor Permissions (GetSensors)

Bit	Value	Description
0	`0x01`	Base (battery)
1	`0x02`	Location (GPS)
2	`0x04`	Environment (temp, humidity, pressure)

Use `0x07` for all permissions.

Sensor Data (Sensors response)

Data returned in CayenneLPP format. See CayenneLPP documentation for parsing.

Cryptographic Algorithms

Operation	Algorithm
Identity / Signing / Verification	Ed25519
Key Exchange	X25519 (ECDH)
Encryption	AES-128 block encryption with zero padding + HMAC-SHA256 (MAC truncated to 2 bytes)
Hashing	SHA-256

Notes

- Data payload limit (255 bytes) matches MeshCore MAX_TRANS_UNIT; no change needed for KISS “1024+ recommended” (that applies to general TNCs, not MeshCore)
- Modem generates identity on first boot (stored in flash)
- All multi-byte values are little-endian unless stated otherwise
- SNR values in RxMeta are multiplied by 4 for 0.25 dB precision
- TxDone is sent as a SetHardware event after each transmission
- Standard KISS clients receive only type 0x00 data frames and can safely ignore all SetHardware (0x06) frames
- See packet_format.md for packet format